Improved Reservoir Characterization of Thin Beds Using an Advanced Deep Learning Approach

Integration of Well Logs, Seismic Data and CWT-Based Feature Learning

Final BTP Report submitted to Indian Institute of Technology Kharagpur

In fulfillment for the award of the degree of

Bachelor of Science in Exploration Geophysics

By

Fazal Rab Abdali (22EX5PE02)

Abdin Shaikh (22EX5PE03)

Under the supervision of

Professor P. N. S. Roy

Department of Geology and Geophysics

Indian Institute of Technology, Kharagpur

DEPARTMENT OF GEOLOGY AND GEOPHYSICS
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
KHARAGPUR - 721302, INDIA

26 April 2026

DECLARATION

We certify that

- The work contained in this report is original and has been done by us under the guidance of our supervisor.
- The work has not been submitted to any other Institute for any degree or diploma.
- We followed the guidelines provided by the Institute when preparing the report.
- We have confirmed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- Whenever we used materials (data, theoretical analysis, figures, and text) from other sources, we gave due credit to them by citing them in the text of the report and giving their details in the references.

Signature of Student 1

Signature of Student 2

Certificate

This is to certify that the project report entitled "Improved Reservoir Characterization of Thin Beds Using an Advanced Deep Learning Approach" submitted by Fazal Rab Abdali (22EX5PE02) & Abdin Shaikh (22EX5PE03) to Indian Institute of Technology Kharagpur, towards fulfilment of requirements for the award of degree of Bachelor of Science in Exploration Geophysics is a record of bonafide project work carried out under my supervision and guidance.

Date: 26 April 2026
Place: Kharagpur

Professor Paresh Nath Singha Roy

Department of Geology and Geophysics
Indian Institute of Technology Kharagpur

Contribution of Team Members

Joint Contribution Overview

This project was carried out as a collaborative effort, where both team members were actively involved in problem understanding, literature review, data handling, and overall system design. The core objective was jointly defined as developing a CWT-based deep learning framework for improved seismic resolution and reservoir characterization. Both members contributed to the integration of multiple components including well log pre-processing, synthetic seismogram generation, CWT feature extraction, neural network training, and interpretation of results, as well as preparation of the final report and presentation.

Fazal Rab Abdali (22EX5PE02)

- Led the development of the synthetic seismogram generation workflow for all four wells
- Implemented depth-to-time conversion using DT and AI logs from LAS files
- Designed and coded the Ricker wavelet convolution pipeline for reflectivity-to-synthetic mapping
- Performed inline–crossline identification for well-to-seismic trace matching using Euclidean distance in CDP coordinates
- Led the development of the CWT-based feature extraction pipeline applied across all four well locations
- Designed the input feature matrix construction combining real, imaginary, and magnitude components of CWT

Abdin Shaikh (22EX5PE03)

- Contributed to seismic data loading, trace extraction, and section visualization from SEG-Y files
- Implemented the fully connected neural network (DNN) architecture and training pipeline
- Performed seismic section interpolation experiments and designed the CWT-based interpolation framework
- Conducted well log pre-processing including median filtering and Savitzky-Golay smoothing
- Contributed to spectral analysis, seismic attribute computation, and interpretation of results
- Played a key role in structuring the methodology and writing the technical sections of the report

Collaborative Work

- Comprehensive literature review and identification of research gaps
- Dataset preparation including loading of SEG-Y seismic data and LAS well log files
- Design of the end-to-end workflow integrating well logs, CWT features, and neural network prediction
- Training, testing, and validation of the deep learning model across multiple well configurations
- Preparation of the final presentation, documentation, and report

Acknowledgements

We would like to express our sincere gratitude to Professor P. N. S. Roy, Department of Geology and Geophysics, IIT Kharagpur, for his constant guidance, encouragement, and constructive feedback throughout this project. His expertise in applied geophysics and seismic interpretation provided invaluable direction at every stage of this work.

We are grateful to the authors of the key research papers that form the scientific foundation of this work - in particular, Manzoor et al. (2024) whose integrated CWT and deep learning approach for thin-bed characterization directly inspired our methodology.

We thank the organizers of the F3 Demo dataset from the Netherlands North Sea for making this high-quality 3D seismic volume and associated well logs freely available to the research community. This dataset, accessible through OpendTect, formed the primary data source for all experiments in this study.

Fazal Rab Abdali (22EX5PE02) | Abdin Shaikh (22EX5PE03)
IIT Kharagpur, 26 April 2026

Abstract

Seismic resolution is fundamentally constrained by the band-limited nature of acquired data, limiting the detection of thin beds that fall below the tuning thickness - typically one-fourth of the dominant wavelength. This study presents an integrated machine learning and Continuous Wavelet Transform (CWT)-based workflow for improved seismic resolution and reservoir characterization, applied to the publicly available F3 Demo 3D seismic dataset from the Netherlands North Sea.

The proposed methodology involves three key stages. First, high-frequency synthetic seismograms are generated at four well locations (F02-1, F03-2, F03-4, and F06-1) using well log data comprising compressional slowness (DT) and acoustic impedance (AI). Compressional velocity is derived from DT, and the acoustic impedance log is converted to the time domain and resampled at 4 ms. Reflection coefficients are then computed and convolved with a zero-phase 70 Hz Ricker wavelet to produce synthetic seismic traces that represent the high-frequency target signal. Second, the real seismic trace at each well location is decomposed using CWT with a complex Morlet wavelet (cmor1.5-1.0) over a frequency range of 5–80 Hz, yielding 100 frequency bands. The real, imaginary, and magnitude components of the CWT are concatenated to form a 300-dimensional input feature vector at each time sample. Third, a fully connected deep neural network (300 → 150 → 75 → 1) is trained to learn the mapping from CWT-derived seismic features to high-frequency synthetic amplitudes, effectively performing spectral enhancement at each time sample.

The model is evaluated under three configurations: training on two wells and testing on the remaining two; training on three wells and testing on one blind well; and training on all four wells for inference on a full 2D inline seismic section. The results demonstrate the model's ability to generalize beyond well locations and produce enhanced seismic predictions over the entire 2D section.

Additionally, this work was preceded by a comprehensive analysis of the F3 dataset during Semester 7, which included well log pre-processing, spectral decomposition, seismic attribute extraction (instantaneous amplitude, phase, and frequency), and CWT-based spectral analysis. These preparatory investigations established the geophysical context and data understanding that informed the deep learning workflow developed in Semester 8. Future work proposes a CWT-based seismic trace interpolation framework, where CWT features from neighbouring traces are used to reconstruct missing or irregularly sampled traces, offering a novel complement to existing CNN-based interpolation methods.

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1. Introduction

1.1 Background and Motivation

Understanding 3D seismic data, assisted by various geophysical techniques, is crucial since it provides significant subsurface information for reservoir characterization. The facies characteristics retrieved from seismic datasets are hindered by numerous constraints such as acquisition geometries, band-limited frequencies, and heterogeneous thin layers that impose limitations on vertical resolution. Improving seismic temporal resolution becomes pivotal especially when seismic traces contain wavelengths greater than a quarter of the reservoir bed thickness, rendering individual reservoir reflections indiscernible.

Conventional exploration strategies targeting major structural reservoirs have largely been exhausted, and the exploration trend has shifted toward subtler, secondary reservoirs characterized by thin beds and complex stratigraphic traps. Conventional interpretation techniques generally become ineffective when dealing with narrow inter-bedded reservoirs that fall below the threshold of seismic resolution. The seismic data is often band-limited due to variables such as the spectral constraint of seismic sources or frequency-dependent wave attenuation during acquisition. As a result, various deep learning approaches have been investigated to address amplitude anomalies caused by thin-bed tuning and to extend the effective resolution of seismic data.

Modern machine learning (ML) algorithms have attracted considerable interest from geophysicists seeking to address thin-bed characterization challenges. Artificial neural networks (ANNs), and in particular deep neural networks (DNNs), have demonstrated the capacity to derive high-level features from unprocessed data through statistical learning over large training sets. The Continuous Wavelet Transform (CWT), on the other hand, provides time-frequency spectra from seismic data, offering superior temporal resolution compared to narrow-band spectral analysis methods. The integration of CWT-derived spectral features with deep learning provides a powerful and cost-effective framework for seismic resolution enhancement.

This study is motivated by the need for a computationally tractable, data-driven approach to seismic enhancement that does not require a large number of wells, detailed stratigraphic models, or full-waveform inversion. By leveraging CWT features of the real seismic trace and mapping them to high-frequency synthetic amplitudes derived from well logs, the proposed framework enables spectral broadening with minimal geological assumptions.

1.2 The F3 Demo Dataset

This study uses the F3 Demo 3D seismic dataset, a publicly available dataset from the Netherlands North Sea, widely used in the geophysical research community for developing and benchmarking seismic interpretation workflows. The dataset includes a full 3D pre-stack time-migrated seismic volume with 600,515 traces and 462 time samples at 4 ms sampling interval, covering a total two-way travel time of approximately 1,848 ms. Four wells are available in the dataset -- F02-1, F03-2, F03-4, and F06-1 -- each containing compressional slowness (DT), bulk density (RHOB), gamma ray (GR), and acoustic impedance (AI) logs in the depth domain.

1.3 Problem Statement

Seismic data is band-limited, and its vertical resolution is fundamentally constrained by the tuning thickness ($\lambda/4$), which for the F3 dataset with a dominant frequency of approximately 30-40 Hz corresponds to approximately 15-20 m. Thin beds that are only a few meters thick commonly lie below this resolution limit and therefore remain undetected in standard seismic interpretation. The objective of this project is to computationally reconstruct high-frequency seismic content by training a deep neural network to map CWT-derived low-frequency spectral features of real seismic traces to high-frequency synthetic seismic traces generated from well logs.

1.4 Objectives

- To preprocess well log data (DT, AI) from four wells in the F3 dataset and perform depth-to-time conversion.
- To generate high-frequency synthetic seismograms at each well location using reflectivity convolution with a Ricker wavelet.
- To identify the inline-crossline location of each well in the 3D seismic volume using CDP coordinate matching.
- To extract CWT features (real, imaginary, and magnitude components) from real seismic traces at well locations.
- To train a fully connected deep neural network to map CWT features of real seismic to high-frequency synthetic amplitudes.
- To evaluate the trained model on blind wells and on full 2D seismic sections for generalization.
- To propose a CWT-based seismic trace interpolation framework as future work for data reconstruction.

2. Literature Review

The problem of seismic resolution enhancement has been studied extensively in the literature. Traditional approaches such as spectral whitening, deconvolution, and model-based seismic inversion have been the mainstay of industry practice for decades. However, these methods require accurate wavelet estimation and rely heavily on assumptions of stationarity that may not hold in complex geological settings.

Manzoor et al. (2024) present an integrated CWT and deep neural network approach for improved reservoir characterization of thin beds, applied to the Lower Indus Basin. In their study, a DNN is trained on CWT-derived spectral features to generate high-frequency acoustic impedance synthetics at the reservoir level, which are then used to enhance seismic resolution. The study demonstrates a significant improvement in thin-bed resolution, reducing the effective tuning thickness from 54 m to 25 m. This work forms the primary methodological reference for the present study.

Zhang et al. (2025) proposed a CWT-based multi-frequency seismic inversion method combined with deep learning to improve thin-layer characterization. By decomposing seismic data into multiple frequency bands, the model captures multi-scale features and predicts acoustic impedance more accurately. The integration of CNN and Transformer architectures further enhances prediction performance and reduces overfitting. This study supports the use of CWT-based spectral features in improving seismic resolution, similar to the approach used in the present work.

Chopra and Marfurt (2015) discussed the application of Continuous Wavelet Transform (CWT) for seismic spectral decomposition, emphasizing that the choice of mother wavelet significantly influences the resolution and interpretability of seismic features. Their study shows that the Morlet wavelet provides more robust and clearer results for identifying geological features such as channels and thin beds compared to other wavelets. This supports the selection of the Morlet wavelet in the present study for effective time–frequency feature extraction.

Mukherjee et al. (2026) propose a 3D U-Net deep learning approach for seismic data recovery, applied to the Amguri prospect in the Upper Assam Shelf of India. Their study demonstrates that convolutional neural networks can effectively reconstruct missing or irregularly sampled seismic traces. The 3D U-Net architecture exploits the spatial continuity of seismic reflectors across inline and crossline directions, yielding high-quality reconstructions with preserved lateral coherency. This work provides the methodological basis for the proposed CWT-based interpolation framework described in the future work section of this report.

The use of complex Morlet wavelets for seismic time-frequency analysis has been well established in the literature. The cmor1.5-1.0 parameterization, with bandwidth parameter 1.5 and center frequency 1.0, provides a balanced trade-off between time and frequency resolution that is suitable for seismic data at typical sampling rates of 4 ms. This parameterization has been validated across multiple seismic datasets and is recommended in the PyWavelets documentation for oscillatory signals.

3. Seismic Data Analysis and Well Log Preprocessing

The work carried out during Semester 7 focused on building a thorough understanding of the F3 Demo dataset through well log preprocessing, seismic visualization, spectral analysis, and seismic attribute computation. This preparatory phase established the geophysical context and data familiarity that underpinned the deep learning workflow developed in Semester 8.

3.1 Well Log Data Description and Preprocessing

The well log data used in this study consist of RHOB (Bulk Density), DT (Sonic Transit Time), GR (Gamma Ray), and AI (Acoustic Impedance), containing 14,064 depth samples across all wells. A statistical summary of the data via the .describe() function reveals realistic values typical of sedimentary formations: the mean bulk density is around 2.15 g/cc, the average sonic transit time is approximately 433 µs/ft, indicating mixed lithologies. Gamma Ray values vary between −3 and 133 API, suggesting alternating clean and shaly zones, while effective porosity (PHIE) ranges from −0.21 to 0.46, with a mean of 0.31, implying moderately porous reservoir intervals.

Before further analysis, the logs were visualized and inspected for outliers and spikes. To improve data quality, a median filter was applied to each log to remove random spikes while preserving edges. A Savitzky–Golay smoothing filter was subsequently applied to obtain continuous, noise-reduced curves suitable for interpretation and attribute computation. These preprocessing steps ensure that the logs are consistent, smooth, and free from acquisition noise, forming a reliable input for depth-to-time conversion and synthetic seismogram generation.

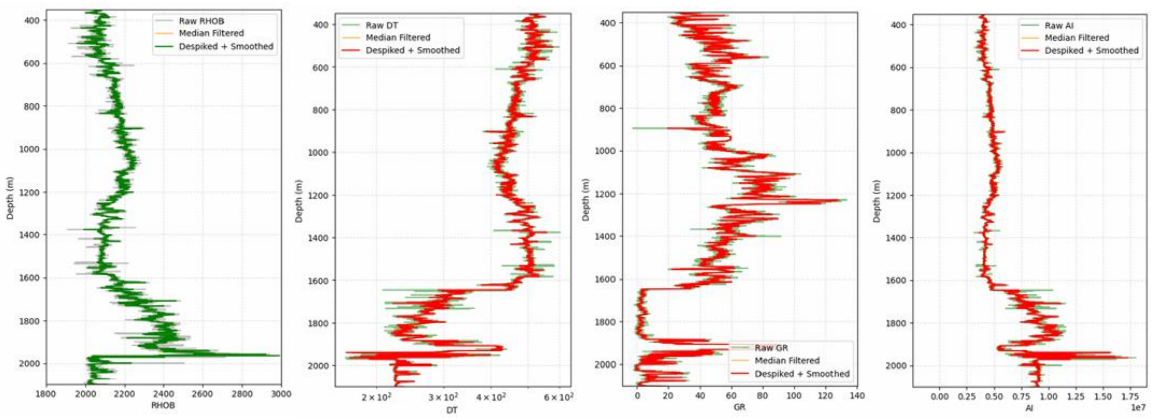

Figure 3.1: Well log preprocessing results showing raw (green), median filtered (orange), and despiked + smoothed (red) curves for RHOB, DT, GR, and AI logs.

3.2 Seismic Data Loading and Visualization

The 3D seismic data was loaded in SEG-Y format using the segyio Python library. The dataset contains 600,515 traces, each with 462 time samples at a 4 ms sampling interval, covering a total two-way travel time of approximately 1,848 ms. To verify correct data reading, a single seismic trace was extracted and plotted to confirm amplitude polarity and sampling characteristics. The amplitude variation shows clear reflection patterns consistent with the expected geological stratigraphy of the Netherlands North Sea.

Inline and crossline section views were generated using imshow() with a red-blue seismic color scale for consistent polarity display. A time-slice at 520 ms was also extracted to examine lateral amplitude variations across the survey area.

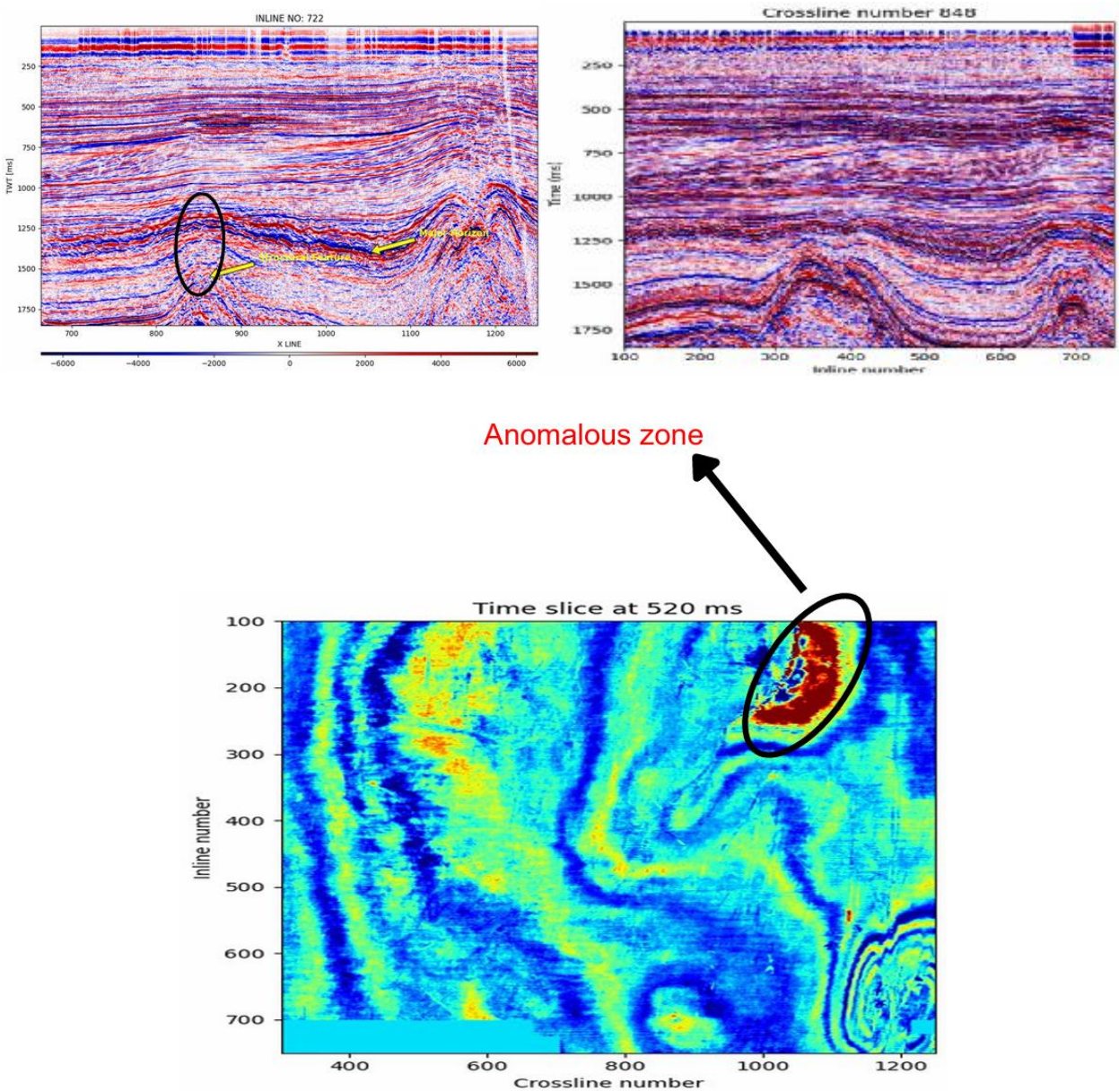

Figure 3.2: Inline, crossline, and time-slice views of the F3 seismic dataset. The inline section (Inline 722) shows major structural features including a prominent horizon. The time-slice at 520 ms reveals an anomalous zone of interest.

3.3 Spectral Analysis using CWT

Continuous Wavelet Transform (CWT) analysis was applied to a representative seismic trace (Inline 722, Crossline 848) to study the time-frequency characteristics of the seismic signal. The complex Morlet wavelet (cmor1.5-1.0) was used over a frequency range of 5 to 100 Hz. The resulting scalogram reveals the real component, imaginary component, and magnitude (energy) of

the seismic signal as a function of time and frequency. The CWT scalogram shows when and where certain frequencies dominate and can help detect thin-bed tuning zones (via frequency shifts), gas-saturated zones (via frequency attenuation), and temporal variation in dominant frequency with depth.

Three distinct spectral anomalies were identified in the CWT magnitude plot: a low-frequency anomaly around 250 ms, a high-frequency anomaly around 500 ms, and a mid-frequency anomaly around 1,250 ms. These anomalies correspond to geologically significant horizons and reflectors in the seismic section.

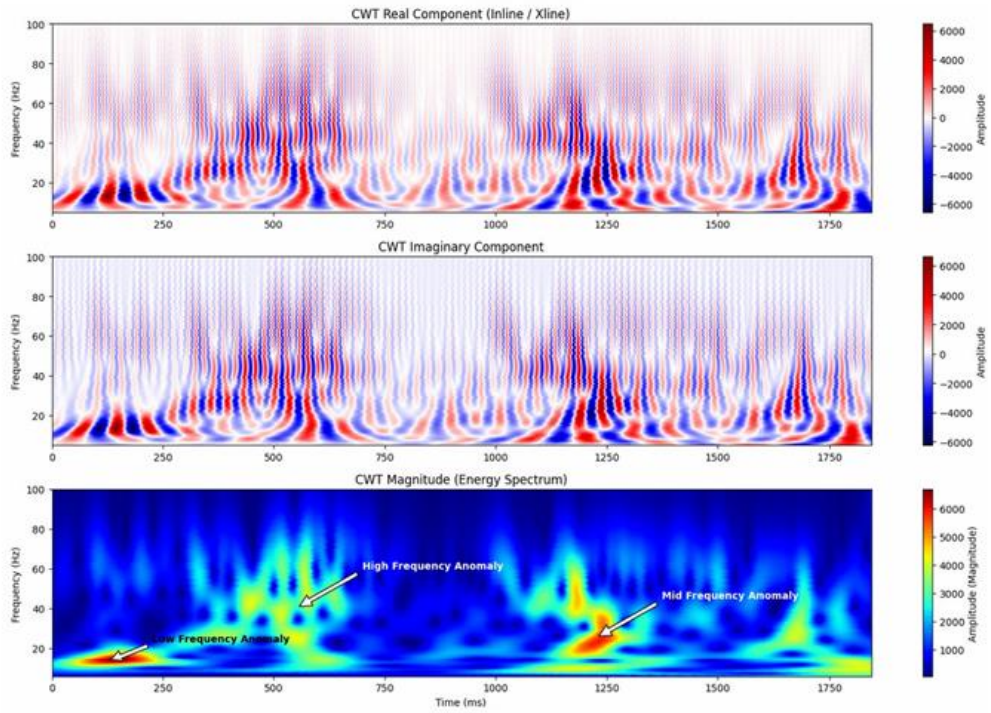

Figure 3.3: CWT scalograms of a representative seismic trace. The magnitude panel (bottom) highlights low-frequency, high-frequency, and mid-frequency anomalous zones that correspond to major subsurface features.

3.4 Seismic Attributes

Several seismic attributes were computed for the F3 dataset to aid in geological interpretation:

Instantaneous Amplitude (Envelope): The envelope of the seismic signal represents the instantaneous energy and is proportional in magnitude to the reflection coefficient. It is useful for highlighting discontinuities, changes in lithology, faults, tuning effects, and sequence boundaries. A high-amplitude anomaly was identified at 520 ms in the instantaneous amplitude time-slice, consistent with the anomalous zone observed in the raw amplitude time-slice.

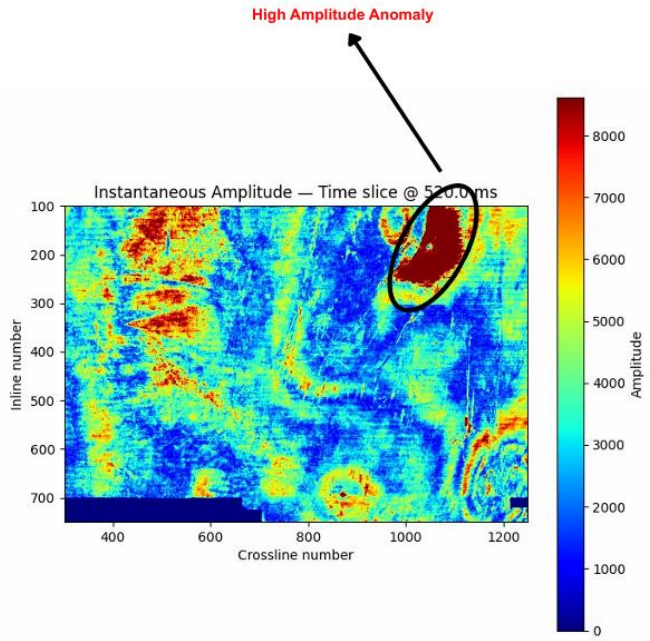

Figure 3.4: Instantaneous amplitude time-slice at 520 ms. The high-amplitude anomaly (circled) indicates a significant acoustic impedance contrast, suggestive of a potential reservoir feature.

Instantaneous Phase: The instantaneous phase, given by $\varphi(t) = \arctan[H(t)/T(t)]$ where $H(t)$ is the Hilbert transform, is measured in degrees $(-\pi, \pi)$. It is independent of amplitude and serves as the best indicator of lateral continuity, sequence boundaries, and bedding configurations.

Instantaneous Frequency: The instantaneous frequency is the time derivative of the instantaneous phase, $F(t) = d\varphi(t)/dt$. It represents the mean amplitude of the seismic wavelet and can indicate bed thickness, lithology parameters, and hydrocarbon presence through low-frequency anomalies. A compact lithology zone associated with a high-frequency region and a frequency anomaly were identified in the instantaneous frequency time-slice at 520 ms.

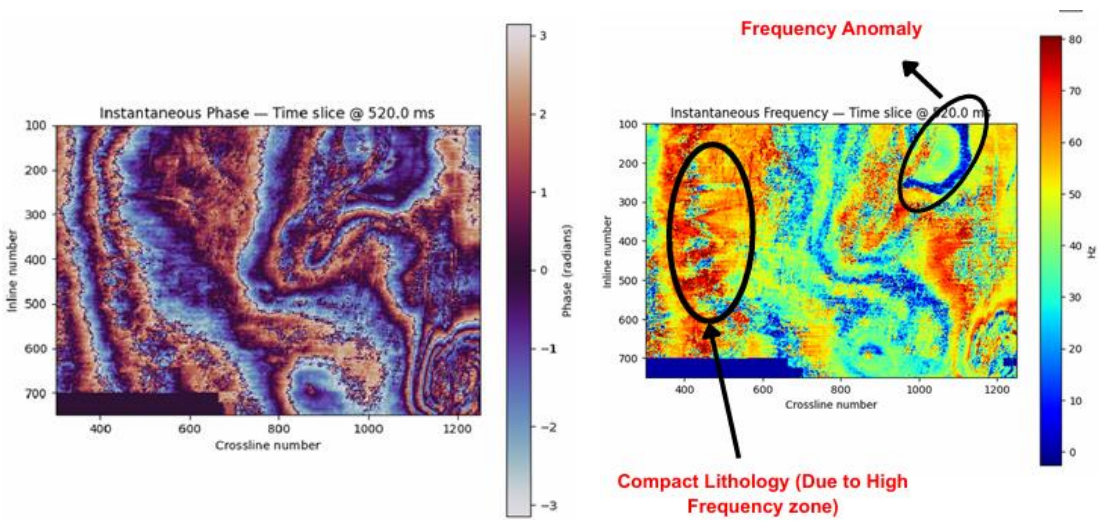

Figure 3.5: Instantaneous phase (left) and instantaneous frequency (right) time-slices at 520 ms. The frequency anomaly and compact lithology zone are annotated.

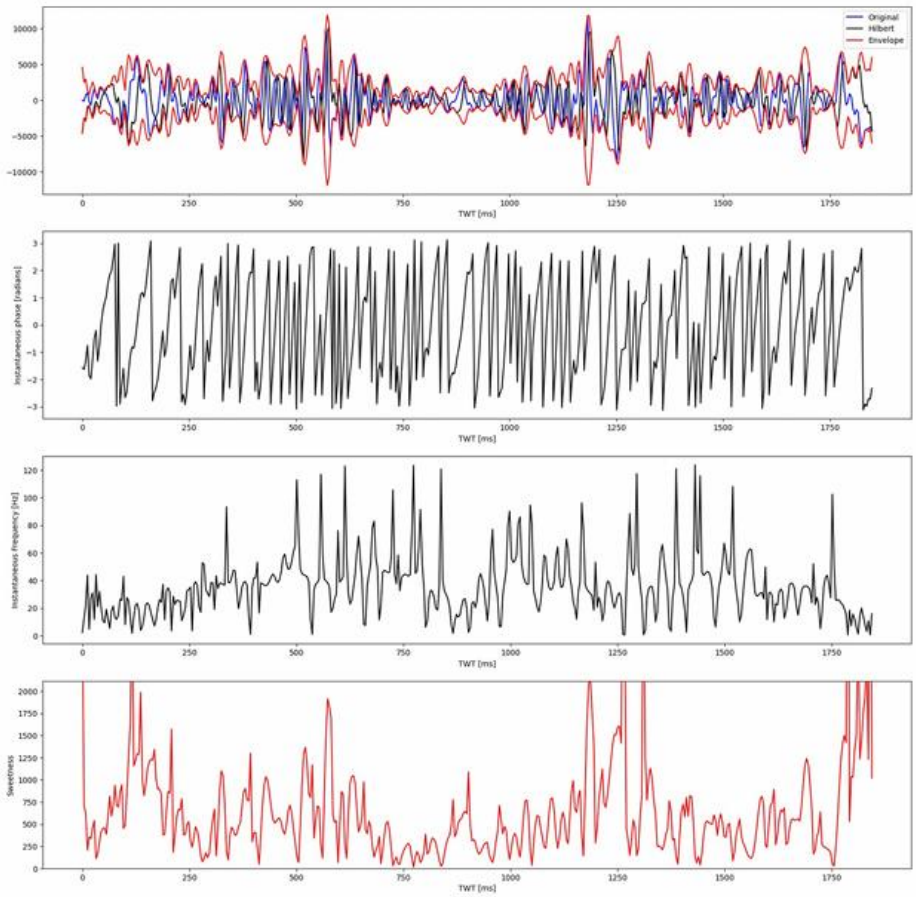

Figure 3.6: Multi-panel seismic attribute display for the trace at Inline 722, Crossline 848, showing the original signal, Hilbert transform, envelope, instantaneous phase, instantaneous frequency, and sweetness as functions of TWT.

3.5 Spectral Decomposition

Spectral decomposition was applied to the F3 seismic dataset to analyze the frequency-dependent response of subsurface reflectors. A 60 Hz spectral band was extracted using a short-time Fourier transform approach with a Hann window of 64 samples (~256 ms) and a bandwidth of ±3 Hz. The resulting spectral magnitude map represents the average reflection strength near 60 Hz at each spatial location (inline × crossline) within the selected time window centered at 520 ms.

The analysis reveals a localized high-magnitude zone surrounded by regions of relatively low spectral energy. This bright zone corresponds to a reflector tuned or resonant with the 60 Hz wavelength, corresponding to an approximate bed thickness of 10–12 m assuming an average velocity of 2,500 m/s. The low-magnitude regions indicate weaker high-frequency energy, which may correspond to thicker beds, softer lithologies, or attenuating layers.

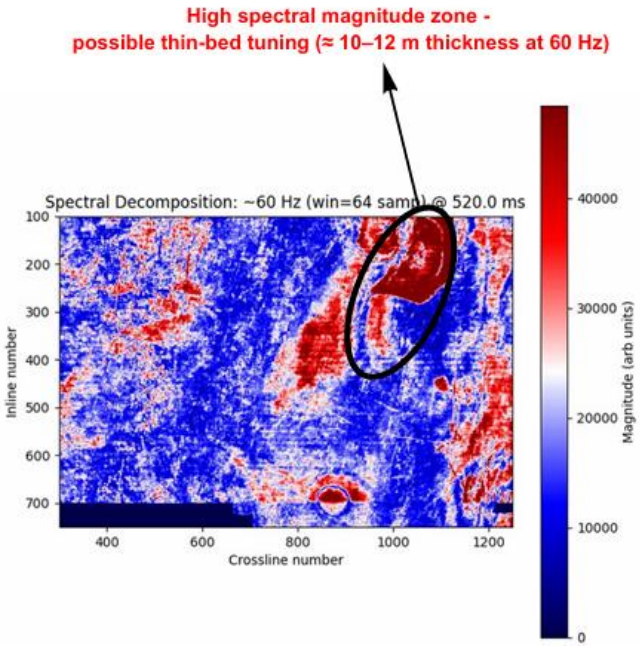

Figure 3.7: Spectral decomposition map at ~60 Hz at 520 ms. The high spectral magnitude zone (circled) indicates possible thin-bed tuning at approximately 10–12 m thickness.

3.6 Summary and Transition to Semester 8

The previous work established a solid understanding of the F3 dataset's geophysical characteristics. Well log pre-processing demonstrated the presence of alternating clean sands and shaly zones across the four wells. The seismic data visualization confirmed the presence of structurally complex horizons and an anomalous high-amplitude zone at 520 ms that may correspond to a potential reservoir. The CWT scalograms and spectral decomposition highlighted frequency-dependent tuning effects and possible thin-bed intervals.

However, owing to the limited number of wells (only four in the entire dataset) and the absence of detailed facies information or geological logs beyond DT and AI, conventional seismic inversion or facies classification approaches could not be meaningfully pursued. Consequently, based on discussions with Professor P. N. S. Roy, the project pivoted in Semester 8 toward a data-driven deep learning framework that could extract maximum information from the available well and seismic data - specifically the CWT-based neural network approach for seismic resolution enhancement, and subsequently a proposed CWT-based interpolation methodology.

4. Methodology (Semester 8)

4.1 Overview of the Proposed Workflow

The new workflow consists of three sequential stages: (1) synthetic seismogram generation from well logs, (2) CWT feature extraction from real seismic traces at well locations, and (3) training a fully connected deep neural network to map CWT features to high-frequency synthetic amplitudes. The overall pipeline is illustrated below:

$$\textit{Well Logs (DT, AI)} \rightarrow \textit{Vp Computation} \rightarrow \textit{Depth-to-Time Conversion} \rightarrow$$
$$\textit{Reflectivity} \rightarrow \textit{Ricker Wavelet Convolution} \rightarrow \textit{Synthetic Seismic}$$
$$+$$
$$\textit{Real Seismic at Well Location} \rightarrow \textit{CWT (Real + Imaginary + Magnitude)} \rightarrow \textit{300-D Feature Vector}$$
$$+$$
$$\textit{DNN (300} \rightarrow \textit{150} \rightarrow \textit{75} \rightarrow \textit{1)} \rightarrow \textit{Predicted High-Frequency Seismic}$$

4.2 Inline-Crossline Identification at Well Locations

To extract the seismic trace corresponding to each well location, the well's X–Y surface coordinates were matched with the seismic survey geometry using CDP-X and CDP-Y coordinates stored in the SEG-Y trace headers. The Euclidean distance between the well location and all seismic CDP locations was computed, and the trace corresponding to the minimum distance was selected. This procedure uniquely identifies the inline–crossline pair associated with each well, enabling accurate extraction of the real seismic trace for CWT feature computation and comparison with the synthetic seismogram.

The resulting inline–crossline pairs and trace indices for all four wells are summarized in Table 4.1.

Well	Inline	Crossline	Trace Index	TWT Start (ms)
F02-1	362	336	249198	50
F03-2	722	848	583984	31.6
F03-4	442	1007	325949	30.66
F06-1	244	387	137031	30.85

Table 4.1: Inline, crossline, trace index, and initial TWT for all four wells in the F3 dataset.

4.3 Synthetic Seismogram Generation

Synthetic seismograms were generated for each well using the following sequential steps:

Step 1 - Depth-to-Time Conversion: The compressional velocity Vp was computed from the DT log using $Vp = 10^6/DT$ (m/s). Two-way travel time (TWT) was then accumulated by cumulative

summation of incremental time steps, where each step equals 2 × Δz/Vp, with Δz = 0.15 m (the depth sampling interval). The initial TWT was set to twice the one-way travel time to the first depth sample, computed as T = 2 × depth/Vp.

As an example, for well F02-1: the first depth point is 48 m with AI = 3,820,257 and DT = 524.05 μs/ft, giving Vp = 1,908.22 m/s and an initial TWT of approximately 50 ms.

Step 2 - AI Resampling: The acoustic impedance log (in the depth domain) was interpolated to the time domain using scipy's interp1d function and resampled to a uniform 4 ms interval. This produces a time-domain AI log suitable for reflectivity computation.

Step 3 - Reflectivity Computation: Reflection coefficients were computed from the time-sampled AI series using the standard impedance contrast formula:

$$R(t) = [AI(t+\Delta t) - AI(t)] / [AI(t+\Delta t) + AI(t)]$$

Step 4 - Ricker Wavelet and Convolution: A zero-phase Ricker wavelet was generated with a dominant frequency of 70 Hz and a wavelet length of 0.12 s at a 4 ms sampling interval. The synthetic seismogram was obtained by convolving the reflectivity series with this wavelet using numpy's convolve function with 'same' mode.

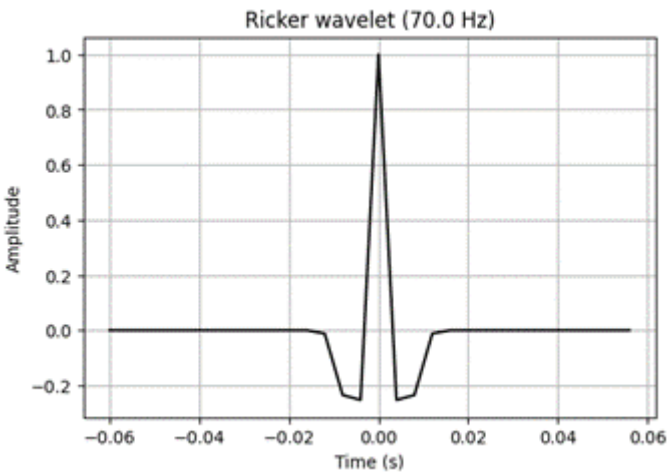

Figure 4.1: Zero-phase Ricker wavelet at 70 Hz.

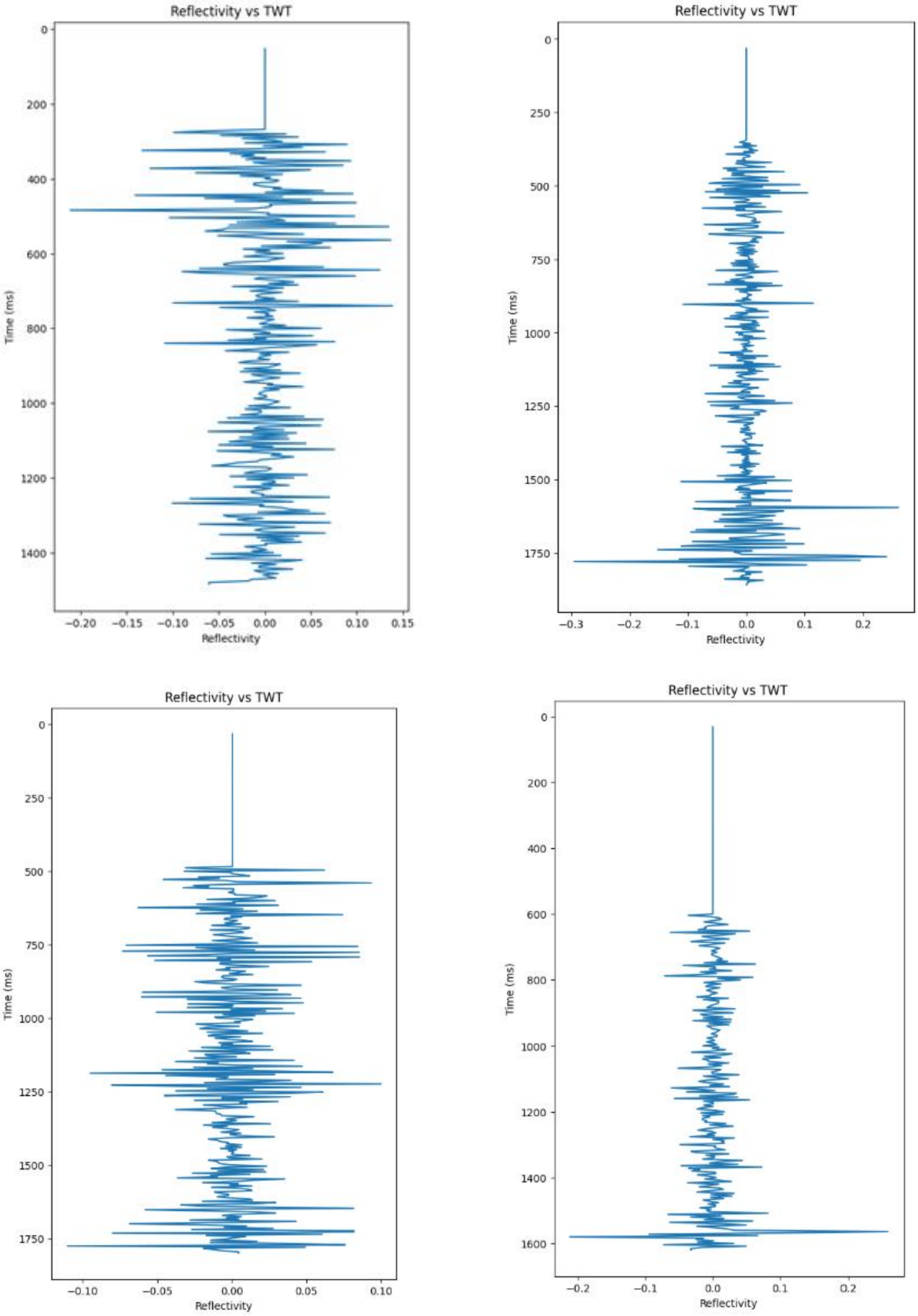

Figure 4.2: Reflectivity series vs. TWT for well F02-1, F03-2, F03-4 and F06-1 respectively, showing acoustic impedance contrasts throughout the logged interval.

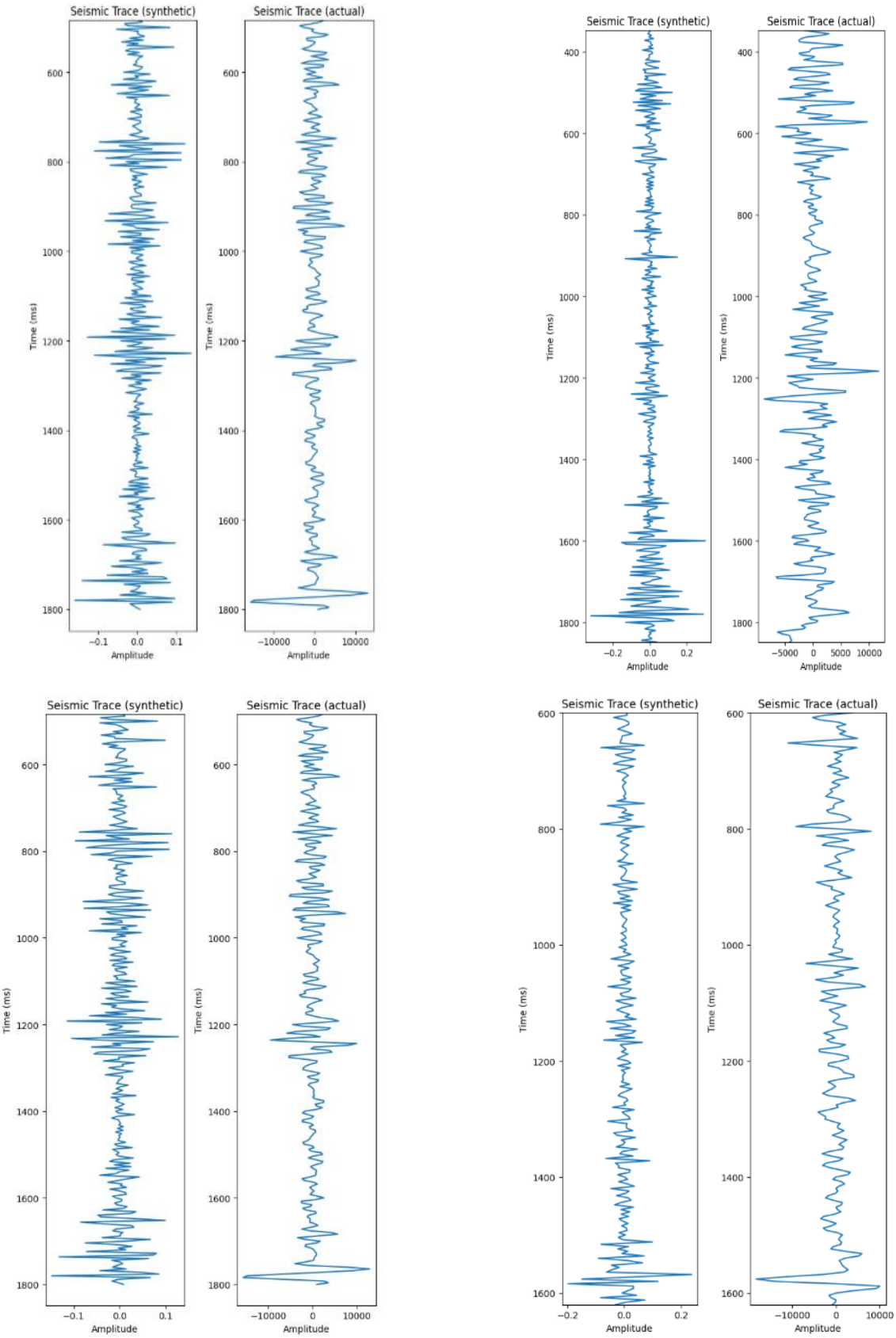

Figure 4.3: Comparison of synthetic seismogram (left) and actual seismic trace (right) at the well location for wells F02-1, F03-2, F03-4 and F06-1 respectively. The morphological similarity between the two traces validates the well-to-seismic tie.

4.4 CWT Feature Extraction

The Continuous Wavelet Transform was applied to the real seismic trace extracted at each well location. The analysis was restricted to the time interval where acoustic impedance log data is available, using a Boolean mask applied to the seismic trace. This ensures strict time alignment between the CWT-derived input features and the synthetic seismic target values.

A complex Morlet wavelet (cmor1.5-1.0) was selected due to its suitability for oscillatory seismic signals and its ability to provide both amplitude and phase information simultaneously. The frequency range was set from 5 Hz to 80 Hz, sampled using 100 logarithmically spaced frequencies to ensure higher resolution at low frequencies where seismic energy and thin-bed effects are most prominent. These frequencies were converted to wavelet scales using the pywt.frequency2scale() function, explicitly accounting for the seismic sampling period of 0.004 s.

The CWT produces complex-valued coefficients as a function of time and frequency. Three components were extracted: the real part, the imaginary part, and the magnitude (|coefs|). For each time sample along the well-log interval, the values across all 100 frequency bands were collected for each component, and the three components were concatenated horizontally to form a 300-dimensional feature vector per time sample (100 real + 100 imaginary + 100 magnitude).

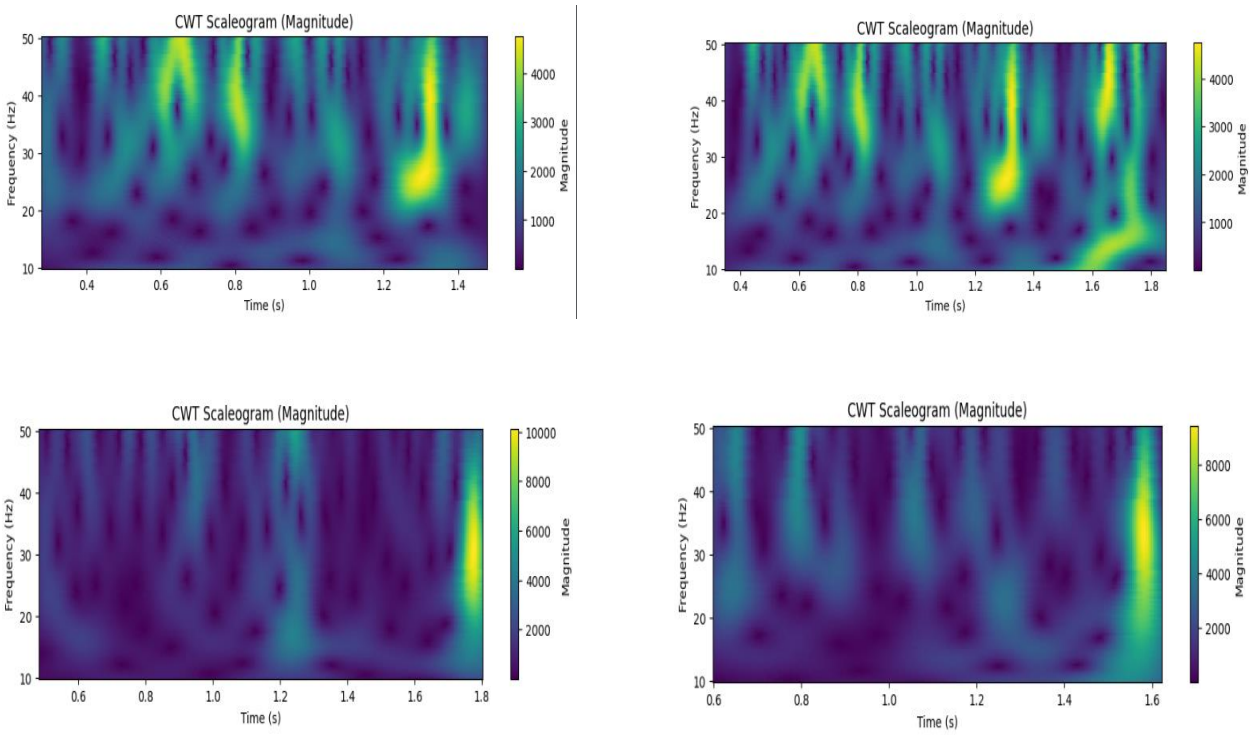

Figure 4.4: CWT magnitude scalograms for the seismic traces at well F02-1, F03-2, F03-4 and F06-1 respectively. Each scalogram shows spectral energy distribution as a function of TWT and frequency, with warm colors indicating high energy.

4.5 Dataset Construction

A tabular dataset was constructed for each well by combining the 300-dimensional CWT feature matrix with the corresponding synthetic seismic amplitude (target) and acoustic impedance (auxiliary) at each time sample. The number of samples per well varies depending on the length of the available log interval: F02-1 contributes 300 samples, F03-2 contributes 376 samples, F03-4 contributes 330 samples, and F06-1 contributes 256 samples, for a total of 1,262 training samples across all four wells. The datasets for individual wells were labelled with their well ID and concatenated into a single combined dataset (final_input.csv) for model training.

4.6 Deep Neural Network Architecture and Training

A fully connected deep neural network (DNN) was implemented using TensorFlow/Keras to learn the nonlinear mapping from 300-dimensional CWT features to the corresponding synthetic seismic amplitude. The network architecture consists of:

- Input layer: 300 neurons (corresponding to the 300-dimensional CWT feature vector)
- Hidden layer 1: 300 neurons, ReLU activation
- Hidden layer 2: 150 neurons, ReLU activation
- Hidden layer 3: 75 neurons, ReLU activation
- Output layer: 1 neuron (predicted synthetic seismic amplitude)

The model was trained using the Adam optimizer with mean squared error (MSE) as the loss function, over 500 epochs with a batch size of 128. Input features were standardized using StandardScaler prior to training. Three training configurations were evaluated:

Case 1: Two wells used for training, remaining two wells used for testing.

Case 2: Three wells used for training, the fourth well used as a test well.

Case 3: All four wells used for training, with testing performed on a full 2D seismic inline section.

These configurations assess the model's generalization capability from sparse well control to a full 2D seismic volume.

5. Results and Discussion

5.1 Training and Validation Loss

The training and validation loss curves were monitored over 500 epochs for all three training configurations. In all cases, both the training loss and validation loss decrease steadily and converge within the first 100–200 epochs, with no significant divergence indicative of overfitting. The use of 20% of training data as a validation split provides a reliable estimate of model generalization within the training well set.

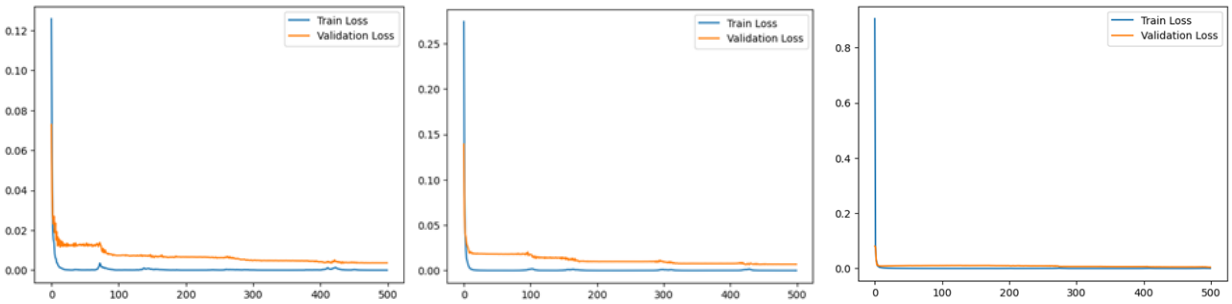

Figure 5.1: Training and validation MSE loss curves for Case 1 (left), Case 2 (middle), and Case 3 (right). Convergence is observed in all cases within approximately 200 epochs.

5.2 Prediction Results: Well Locations

The predicted synthetic seismic traces from the DNN were compared against the true synthetic traces at the test well locations. The predicted and actual traces are plotted as amplitude vs. sample index for each test well. The model captures the broad amplitude envelope and the polarity of major reflections. The predictions exhibit a degree of smoothing relative to the true synthetic trace, which is expected given the finite number of training samples and the regression nature of the task.

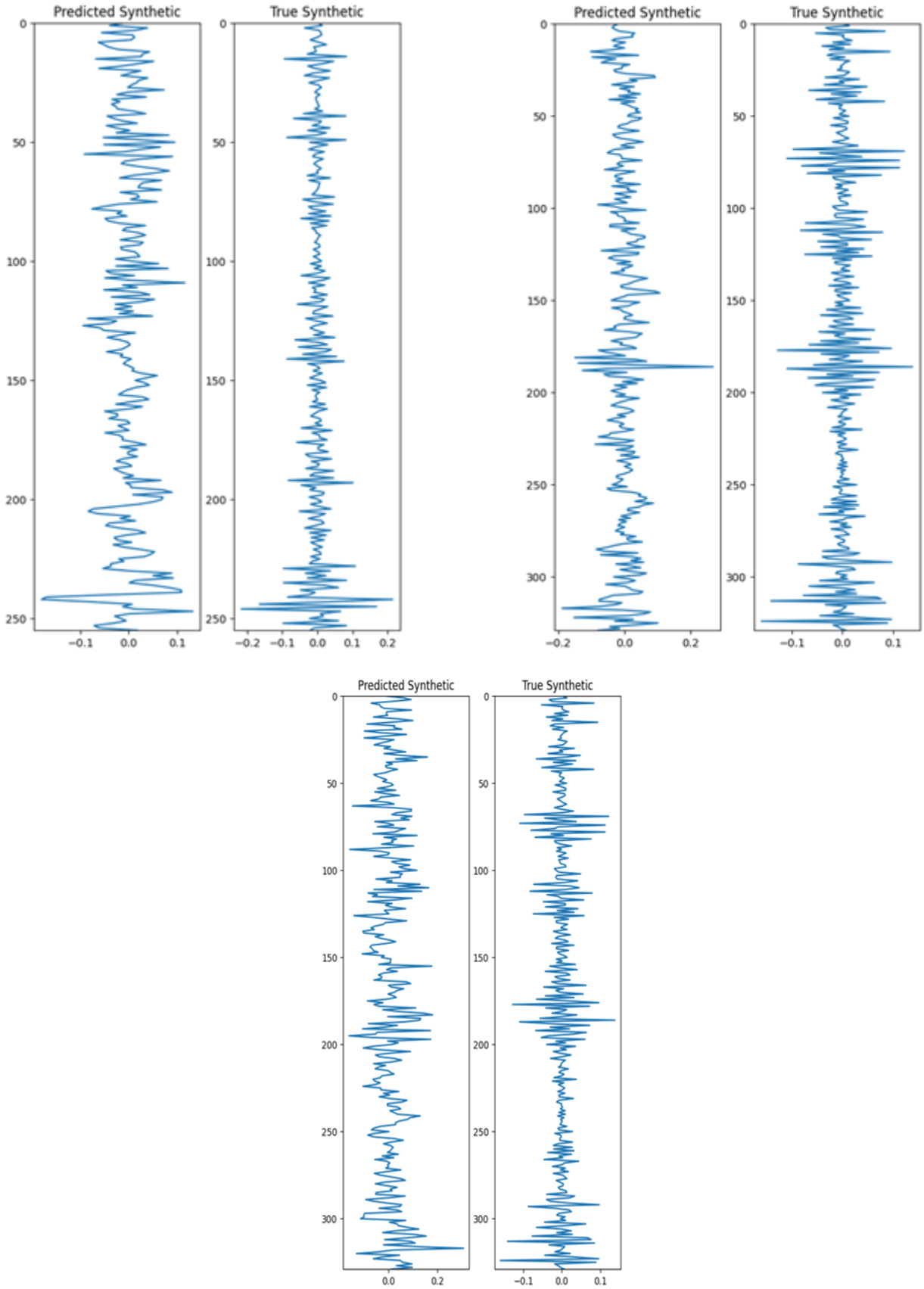

Figure 5.2: Predicted synthetic seismic amplitude (left) vs. true synthetic amplitude (right) at test well locations for Case 1(top) and Case 2(bottom).

5.3 Prediction Results: 2D Seismic Section

In Case 3, the model trained on all four wells was applied to predict the high-frequency synthetic amplitude at every trace of a full 2D inline section (Inline 442 of the F3 dataset). The input CWT features were computed for each trace in the inline, and the trained DNN was used to generate a predicted enhanced seismic section. This demonstrates the model's ability to generalize spatially from well-based training to a full seismic section, which is the practical end goal of the proposed workflow.

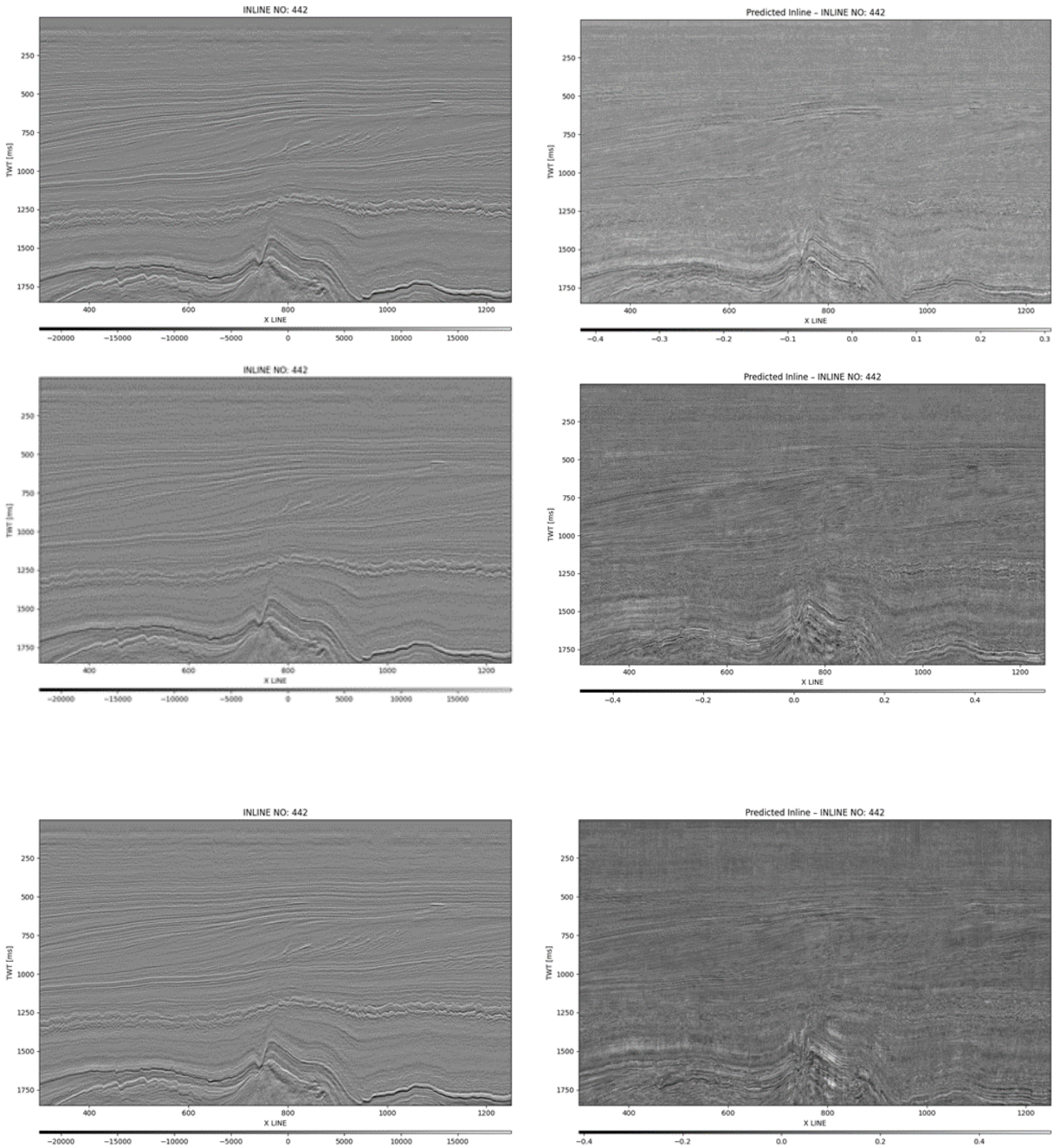

Figure 5.3: Original seismic inline section (Inline 442) and DNN-predicted enhanced seismic section(right) for Case 1(top), Case 2(middle) and Case 3(bottom). The predicted section shows the expected frequency enhancement with preserved lateral continuity.

6. Limitations

Despite the promising results, several limitations of the current study must be acknowledged:

- Limited well control: Only four wells are available in the F3 Demo dataset. With such a small number of training samples (approximately 1,262 total), the model's generalization to the full 3D volume may be restricted. A larger number of wells with geographically distributed coverage would be required for robust production-scale application.
- Absence of facies and geological information: The F3 dataset does not include facies logs or detailed lithological descriptions. Consequently, the model cannot be conditioned on geological context, which limits its ability to distinguish between lithologically distinct thin layers.
- No frequency quantification of enhancement: While the predicted seismic section shows qualitative improvement, a rigorous quantification of the frequency bandwidth enhancement (e.g., through spectral ratio analysis or wedge model tuning tests) was not performed in this study due to the absence of a ground-truth high-resolution seismic section.
- Single inline testing: The 2D seismic testing was limited to a single inline section. Extension to the full 3D volume and evaluation of spatial consistency across crosslines remains as future work.
- No uncertainty quantification: The current DNN framework provides a deterministic point prediction without any estimate of prediction uncertainty. Bayesian neural networks or ensemble approaches could be incorporated in future work to provide confidence bounds on the predictions.

7. Conclusion

This study presents a two-semester investigation into seismic data analysis and deep learning-based reservoir characterization using the publicly available F3 Demo 3D seismic dataset from the Netherlands North Sea.

In Semester 7, a thorough exploratory analysis of the F3 dataset was conducted. Well logs from four wells (F02-1, F03-2, F03-4, F06-1) were preprocessed using median and Savitzky–Golay filters to remove acquisition noise. The 3D seismic volume was loaded and visualized through inline, crossline, and time-slice views, revealing a prominent high-amplitude anomaly at 520 ms. CWT-based spectral analysis identified distinct frequency anomalies at multiple time intervals. Seismic attributes including instantaneous amplitude, phase, and frequency were computed and interpreted. Spectral decomposition at 60 Hz highlighted a possible thin-bed tuning zone of approximately 10–12 m thickness.

In Semester 8, a novel end-to-end deep learning workflow was developed and implemented. High-frequency synthetic seismograms were generated at all four well locations through depth-to-time conversion, reflectivity computation, and Ricker wavelet convolution at 70 Hz. Real seismic traces at the well locations were decomposed using the complex Morlet CWT over 100 frequency bands (5–80 Hz), and the real, imaginary, and magnitude components were concatenated to form 300-dimensional feature vectors. A fully connected DNN (300 → 150 → 75 → 1) was trained to map these features to synthetic seismic amplitudes, and tested under three training configurations including blind-well tests and full 2D inline prediction.

The results demonstrate that the CWT-DNN framework can learn to predict high-frequency synthetic amplitudes from band-limited seismic features, and generalizes spatially to produce enhanced seismic predictions over a full 2D section. The primary limitation of the current work is the small number of available wells, which restricts the diversity and volume of training data.

Future work proposes a CWT-based seismic trace interpolation framework as a complementary approach to CNN-based methods, leveraging the time-frequency coherency of adjacent traces for reconstruction of missing seismic data. Both approaches will be compared quantitatively and are intended to form the basis of a research publication.

The complete code for this project are available at: Code

8. Future Work: CWT-Based Seismic Interpolation

Based on discussions with Professor P. N. S. Roy at the conclusion of this work, two promising directions for future research were identified, both relating to seismic data recovery and interpolation using deep learning.

The first direction is CNN-based seismic interpolation, building on the work of Mukherjee et al. (2026), who demonstrated the use of a 3D U-Net architecture for seismic data recovery at the Amguri prospect. The proposed extension would augment the existing CNN-based workflow by incorporating additional physically motivated features or loss functions that improve reconstruction quality, particularly in structurally complex zones.

The second - and primary direction proposed in this study is a novel CWT-based seismic trace interpolation framework. The approach is motivated by the observation that adjacent seismic traces share strong spatial coherency in their time-frequency characteristics, which can be exploited for trace reconstruction. The proposed workflow is as follows:

Training stage: Given a set of N consecutive seismic traces, randomly designate one trace as the target (missing) trace. Compute CWT features (real, imaginary, and magnitude components) for the two traces immediately to the left and the two traces immediately to the right of the target trace. Concatenate these four CWT feature matrices to form a high-dimensional input feature set. The target output is the original amplitude of the missing seismic trace at each time sample.

Testing stage: For a given set of traces with one or more missing traces (obtained, for example, by randomly removing 30% of traces from an inline section), compute the CWT features of the neighbouring available traces and use the trained model to predict the missing trace amplitudes. Compare the predicted trace against the original using metrics such as RMSE, R^2, MSE, and SSIM.

This approach is conceptually analogous to the U-Net based recovery method but operates in the time-frequency domain rather than the space-time domain. The CWT-based representation may offer advantages in preserving spectral fidelity of the reconstructed trace, particularly for thin-bed tuning effects that manifest as localized spectral anomalies. The two methods (CNN-based and CWT-based) would ultimately be compared and contrasted in a joint publication.

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